

MODEL QUESTION
FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING / TECHNOLOGY

DESIGN OF MACHINE ELEMENTS
TED (15) - 5021

Time : 3 hours

(Maximum marks :100)

PART - A

(Maximum marks :10)

Marks

I. Answer all the questions in one or two sentences. Each question carries 2 marks.

1. Define pitch of a thread?
2. Write the classification of machine design?
3. Define axle?
4. What is the function of a bearing?
5. Name the different belt materials?

(5x2= 10)

PART - B

(Maximum marks : 30)

II. Answer any five of the following questions. Each question carries 6 marks.

1. Write the general procedure in designing a machine?
2. A machine weighing 30 kN is provided with a steel eye bolt for lifting it. If the ultimate tensile strength of steel is 500 MPa and the factor of safety is 6. Find the size of bolt?
3. Briefly describe about different types of couplings?
4. A solid circular transmission shafts of 100 mm diameter is subjected to a torque of 10 kN-m. Calculate the maximum shear stress produced in the shaft?
5. Write six advantages of rolling contact bearings compared with sliding contact bearings?
6. Briefly describe about different types of couplings?
7. Write the advantages of gear drive?

(5x6= 30)

PART – C

(Maximum marks :60)

(Answer one full question from each unit. Each full question carries 15 marks.)

MODULE – I

a. A generator weighing 15 kN is to be provided with an eye bolt in the housing for lifting purposes. Find the size of the bolt if it is made of C-40 steel. If the ultimate tensile strength of C-40 steel is 600 MPa and the factor of safety is 6?

9

b. Design a sunk key for a shaft of 60 mm diameter. The permissible shear stress is 35 MPa and compression stress is 70 MPa. Assume key and shaft are made of same material?

6

OR

- III. a. A screw jack has a double square threaded screw of mean diameter 50 mm and pitch 10 mm. The coefficient of friction between the screw and nut is 0.1, neglecting the collar friction. Calculate the torque required to be applied on the screw for lifting a load of 25 kN?

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- b. A cylinder has an effective diameter 300 mm. It is subjected to a maximum steam pressure of 1.5 MPa. Cylinder cover is fixed by means of 16 studs. Permissible tensile stress in the studs is limited to 30 MPa. Determine the size of studs?

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MODULE –II

- V. a. A steel spindle transmits 1 MW power at 240 rpm. The angular deflection should not exceed 1° per metre of the spindle. If the modulus of rigidity for the material of the spindle is 80 MPa.

Find the diameter of the spindle and shear stress induced in the spindle?

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- b. Derive an expression to find the diameter of a hollow shaft which is subjected to a torque T and the ratio between inner and outer diameter is k ?

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OR

- VI. a. Design a sleeve or muff coupling for a shaft transmitting 40 kW at 140 rpm. The allowable shear and crushing stresses for the shaft and key which are made of mild steel are 42 MPa and 84 MPa respectively. The material of the muff is cast iron with a permissible shear strength of 14 MPa. Assume that the maximum torque transmitted is equal to the mean torque?

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- b. What are the different purposes of couplings?

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MODULE – III

- VII. a. Construct a cam profile using the following data.

(a) Out stroke during 90° of the rotation.

(b) Dwell for the next 30° of rotation.

(c) Return stroke for next 90° .

(d) Dwell for the rest of motion.

Stroke of follower is 25 mm, least radius of cam is 50 mm, diameter of roller is 20 mm.

Axis of the roller follower is offset by 20 mm from the axis of cam shaft. The follower moves with uniform velocity?

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- b. A journal bearing having a diameter of 65 mm is subjected to a load of 5 kN at a speed of 200 rpm. If the length to diameter ratio is 3 and coefficient of friction is 0.02. Find the bearing pressure and heat generated?

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OR

VIII. a. Draw the profile of the cam to give the following motion to a knife edged follower:

- (a) Out stroke during 60° of cam rotation.
- (b) Dwell for the next 30° of cam rotation.
- (c) Return stroke during next 60° of cam rotation.
- (d) Dwell for the remaining 210° of cam rotation.

The minimum radius of cam is 50 mm and the stroke is 30 mm. Axis of the follower is passing through the centre of cam shaft and both strokes of the follower is executed

with SHM?

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b. A vertical shaft 150 mm diameter rotating at 100 rpm rests on a flat end foot step bearing.

The shaft carries a vertical load of 15 kN. If the coefficient of friction is 0.05. Estimate the power lost in friction at the bearing assuming uniform pressure?

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MODULE –IV

IX. a. A 300 mm diameter pulley is fixed on a prime mover shaft and the driven shaft has a pulley of 200 mm diameter. The distance between the both pulleys is 5.3 m apart. Find out the required length of belt for an open belt drive?

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b. Write the advantages of gear drive?

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OR

X. a. Power is being transmitted from one shaft to another by means of a crossed belt. The diameter of larger pulley is 250 mm and that of smaller pulley is 200 mm. The distance between the centres of two pulley is 2.5 m. Find the length of belt. The two shafts are in the same plane and parallel to each other?

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b. Briefly describe about epicyclic gear train?

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